

KHOVD, Mongolia

WMO: 442180

Lat: 47.9958N

Lon: 91.6325E

Elev: 4610

StdP: 12.41

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-26.2	-35.8	0.8	-30.0	-30.9	1.1	-25.1	20.1	16.1	14.6	16.9	2.0	180	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	85.7	59.9	82.7	59.1	80.0	57.9	64.4	78.6	62.1	76.9	60.3	75.1	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	91.6	69.2	56.8	81.8	67.8	54.4	74.7	67.5	32.2	79.1	30.3	77.1	28.9	75.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	16.7	13.2	-29.9	91.2	7.4	3.8	-35.2	93.9	-39.5	96.2	-43.6	98.3	-49.0	101.1		
(5)			-30.1	65.5	7.3	3.9	-35.3	68.3	-39.6	70.6	-43.7	72.8	-49.0	75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	-8.1	0.9	23.4	42.6	54.2	65.2	68.7	64.5	52.6	37.8	17.8	0.0		
	DBStd	27.97	10.46	12.93	12.21	9.55	8.32	6.27	5.27	6.10	8.05	9.01	10.92	10.25		
	HDD50	7282	1803	1376	825	252	56	3	0	1	64	385	968	1549		
	HDD65	11195	2268	1796	1289	672	342	71	25	83	376	842	1417	2014		
	CDD50	1857	0	0	1	31	186	457	579	452	143	8	0	0		
	CDD65	296	0	0	0	1	8	76	139	68	4	0	0	0		
	CDH74	2683	0	0	0	6	99	679	1185	644	70	0	0	0		
	CDH80	605	0	0	0	0	11	139	307	141	7	0	0	0		
	WSAvg	3.7	1.9	2.5	3.9	5.4	5.6	4.8	4.1	3.7	4.0	3.6	3.0	2.2		
	PrecAvg	4.4	0.0	0.0	0.1	0.3	0.5	0.9	1.3	0.9	0.3	0.1	0.1	0.1		
(15) Precipitation	PrecMax	8.9	0.3	0.3	0.6	1.3	2.0	2.6	4.7	4.1	1.1	0.7	0.4	0.3		
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.4	0.1	0.1	0.1	0.2	0.5	0.7	0.9	0.8	0.3	0.2	0.1	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	38.4	58.6	72.5	81.5	87.8	91.4	88.1	80.1	66.7	54.7	39.2		
		MCWB	21.2	29.0	42.8	49.5	57.0	58.9	64.8	60.4	55.8	48.4	40.9	31.0		
	2%	DB	18.5	31.6	53.0	67.1	76.5	83.7	86.5	83.8	75.5	60.9	46.2	28.8		
		MCWB	14.2	24.5	38.6	46.8	53.7	58.7	61.5	58.7	54.4	44.7	35.8	22.7		
	5%	DB	11.6	25.5	48.2	62.9	73.1	80.7	83.4	80.5	71.7	57.0	40.0	20.9		
		MCWB	9.0	19.9	35.6	44.9	52.1	57.6	59.9	58.3	52.4	42.7	31.2	16.9		
	10%	DB	6.2	19.7	43.2	59.0	69.7	77.9	80.6	77.4	67.7	53.4	34.3	14.7		
		MCWB	4.5	15.8	32.7	42.9	50.0	56.5	58.8	57.1	50.7	40.5	27.5	12.0		
	0.4%	WB	21.6	29.5	43.6	52.2	59.6	65.2	70.2	66.1	59.8	49.9	41.7	30.8		
		MCDB	27.7	37.2	57.9	69.5	78.1	81.3	85.9	77.3	75.4	63.9	54.7	38.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	14.3	24.6	39.3	48.1	56.1	62.0	65.5	62.8	56.5	46.0	36.2	22.9		
		MCDB	18.5	31.2	51.5	64.3	73.0	78.5	78.5	78.1	71.4	58.9	45.6	28.1		
	5%	WB	9.1	19.9	36.2	45.7	53.5	59.9	63.2	60.6	53.8	43.5	31.6	17.0		
		MCDB	11.6	25.1	47.5	61.1	70.0	75.4	77.0	75.6	68.8	56.0	39.3	20.9		
	10%	WB	4.5	15.8	32.5	43.4	51.2	58.2	61.2	58.8	51.5	40.9	27.7	12.1		
		MCDB	6.1	19.6	42.4	57.8	67.8	73.5	75.2	73.0	66.2	52.5	34.3	14.6		
	5% DB	MDBR	18.2	21.0	23.1	22.3	21.3	19.7	19.0	19.9	22.4	21.9	19.6	17.5		
		MCDBR	22.4	25.1	26.6	26.0	25.6	23.8	22.8	24.1	26.8	25.9	24.0	23.1		
		MCWBR	19.4	19.6	17.2	14.3	13.1	9.9	9.0	10.1	13.3	15.4	16.6	18.7		
		MCWBR	22.2	24.8	25.7	24.0	23.4	20.8	19.5	21.1	24.4	25.4	23.9	22.8		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.270	0.315	0.360	0.357	0.367	0.363	0.341	0.305	0.286	0.259	0.240		
		taud	2.350	2.276	2.228	2.199	2.268	2.327	2.371	2.405	2.456	2.432	2.364	2.315		
	Ebn at Noon		273	284	285	282	286	284	283	285	288	276	260	255		
		Edh at Noon	22	30	37	42	41	39	37	34	29	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		546	900	1374	1752	2055	2107	2065	1843	1507	953	584	430		
	RadStd		28	39	69	77	135	119	119	67	64	67	39	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	-4.21	N/A	-2.23	-2.81	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	-1.35	-2.11	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page